CMSE: Cross-Modal Semantic Enhancement Network for Classification of Hyperspectral and LiDAR Data

Wenqi Han, Wang Miao, Jie Geng, *Member, IEEE*, and Wen Jiang

***Abstract*— The fusion of hyperspectral image (HSI) and light detection and ranging (LiDAR) data is widely used for land cover classification. However, due to different imaging mechanisms, HSI and LiDAR data always present significant image differences, and the dimensions and feature distributions of HSI and LiDAR are highly dissimilar. This makes it challenging to represent and correlate semantic information from multimodal data. Current methods for classifying pixel-by-pixel features, which rely on cascaded or attention-based fusion, cannot effectively use multimodal features. To achieve accurate classification results, extracting and fusing similar high-order semantic information and complementary discriminative information contained in multimodal data is vital. In this article, we propose a cross-modal semantic enhancement network (CMSE) for multimodal semantic information mining and fusion. Our proposed CMSE framework extracts features from the image on multiple scales, capturing more representative local sparse features with different sizes of convolution kernels. To represent high-level semantic features related to land cover, we establish a Gaussian-weighted matrix and semantically transform the spatial and spectral features of distinct branches. Finally, we build a multilevel residual fusion module to incrementally fuse spectral features from HSI and elevation features from LiDAR. Additionally, we introduce a cross-modal semantically constrained loss to guide multimodal semantic feature alignment. We evaluate our approach on three multimodal remote sensing (RS) datasets, namely the Houston2013, Trento, and MUUFL datasets. The experimental results demonstrate that our proposed CMSE model achieves superior performance in terms of accuracy and robustness compared to other related deep networks.**

***Index Terms*— Classification, land cover, multimodal, remote sensing (RS), semantic features.**

I. Introduction

IN RECENT years, the proliferation of remote sensing (RS) technology has led to the increasing utilization of RS data in various applications [1], [2], [3], [4], [5]. Early research on classifying RS data primarily relied on a singular data type. For example, hyperspectral image (HSI) classification uses solely HSI data [6], [7]. Present HSI classification techniques can yield satisfactory outcomes [8]. Nonetheless, they may occasionally produce incorrect classification, specifically in urban surroundings where buildings' roads and roofing materials may be akin and exhibit comparable spectral curves [9], [10]. Thus, distinguishing these land-cover types using only spectral data proves challenging. However, light detection and ranging (LiDAR) data can conveniently determine the elevation of these two objects, making it easier to distinguish between them. An effective combination of these two modalities can compensate for each other's limitations, thereby improving classification accuracy. Nevertheless, the differences in data structure between HSI and LiDAR data pose a significant challenge for direct classification models, requiring larger amounts of training data to achieve satisfactory performance. To address this challenge, numerous deep learning approaches [1], [11], [12], [13], [14] have been developed to enhance the performance of multimodal RS classification for land cover.

Feature fusion is the primary method used for multimodal fusion in this context, as evidenced by research on deep learning-based approaches [15], [16], [17], [18], [19]. It can be classified into three types based on the fusion phase: early, middle, and late fusion [20]. The research conducted by Hong et al. [20] and Wang et al. [21] has suggested that the early and late fusion methods may limit the interaction of information within and between modalities. Due to its flexibility in accommodating deep learning, intermediate fusion has become the preferred method of multimodal fusion in numerous studies [22]. Recent studies have introduced multimodal image reconstruction as a technique for achieving feature fusion [23], [24], [25]. To further improve multimodal feature interaction, unsupervised CNN models based on encoder–decoder architectures are proposed, and data reconstruction strategies are shown to better activate neurons across modalities [24], [26]. Despite these efforts, these methods use shallow feature classification models that have difficulty handling complex and highly nonlinear sample data distributions.

Most joint classification models for HSI and LiDAR data use the patch-to-patch approach, relying on center pixel classification. In complex multicategory images, more than relying on the center pixel alone may be required to distinguish similar categories, causing confusion and requiring additional information. Efficient representation and interaction of multimodal features remain essential research areas for classifying hyperspectral and LiDAR data. Utilizing the higher-order features of each data source is a critical component of achieving this interaction. Guo et al. [27] regulated the structural dissimilarity between clean and degraded

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The authors are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: hanwenqinwpu@mail.nwpu.edu.cn; mw0638@mail.nwpu.edu.cn; gengjie@nwpu.edu.cn; jiangwen@nwpu.edu.cn).

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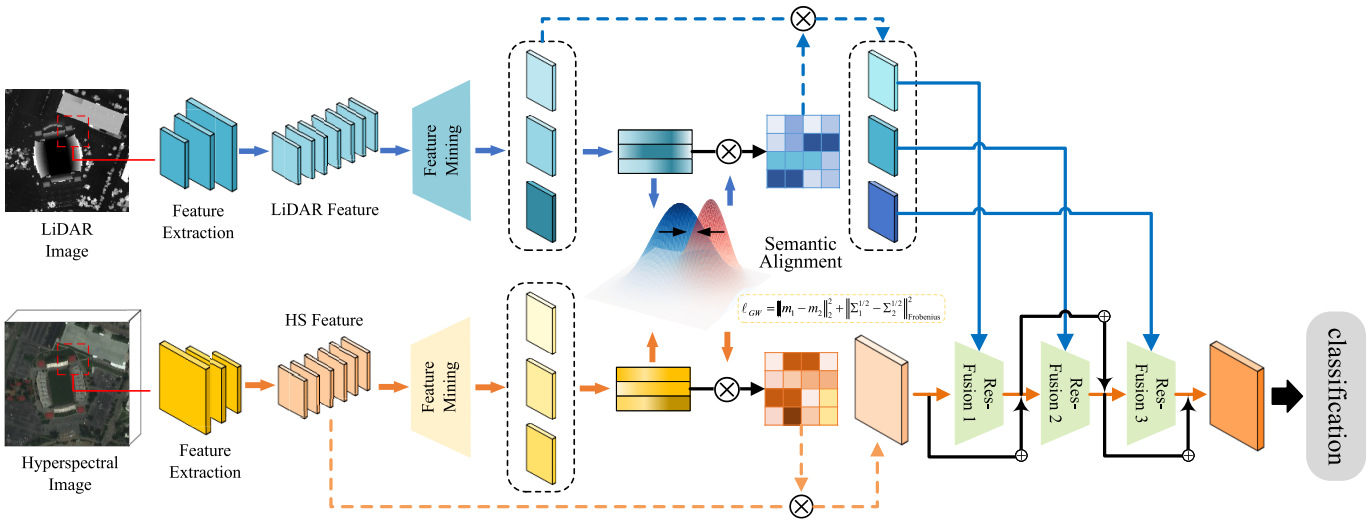

Fig. 1. Our framework of CMSE network for classification of hyperspectral and LiDAR data.

images by exploiting the second-order interaction of the Gram matrix. Meanwhile, matching neural activations based on the Gram matrix is a practical approach for aligning heterogeneous distributions across images [28]. All approaches described above are based on the convolutional neural network (CNN) and related techniques. Although these approaches notably enhance the classification accuracy of HSI and LiDAR data, their implementation is often challenging due to issues such as the insufficient availability of training data and the high computational cost resulting from abundant feature redundancy. Furthermore, while the CNN model can effectively represent spatial features, it needs help managing the massive spectral sequence information embedded in HSI data, which causes difficulties in distinguishing minor spectral differences between adjacent pixels.

To address the previously mentioned challenges, we propose a cross-modal semantic enhancement (CMSE) network to mine and fuse multimodal semantic information. The purpose of our method is to decrease the semantic feature differences between modalities and enable efficient multimodal feature fusion. In the suggested framework, we utilize the CMSE module to extract features from the image at numerous scales by extracting more representative sparse local features via convolutions with different kernel sizes. Next, we create a semantic enhancement module that employs a Gaussian-weighted matrix to represent high-level semantic features that pertain to the land cover and semantically modify the spatial and spectral features associated with distinct branches. It is employed to supplement the high-level characteristics of HSI and LiDAR. Finally, we devise a multilevel residual fusion module that compensates for the inadequate elevation information in the spectral features by performing a step-by-step fusion approach. The framework of our CMSE network for hyperspectral and LiDAR data classification is shown in Fig. 1. In summary, the noteworthy contributions of our work are as follows.

1) We present a novel hyperspectral and LiDAR data classification approach called CMSE. The CMSE not only extracts features at multiple scales from the images but also enhances the interaction between cross-modal features, thereby facilitating the generation of consistent representations across modalities. Furthermore, the CMSE employs a multilevel residual fusion strategy for hyperspectral and LiDAR data, which results in a comprehensive and robust data representation.
2) We propose a semantic enhancement module based on a Gaussian-weighted matrix to tackle the issues of interclass confusion and inaccurate representation of multimodal data in classification tasks. The module utilizes shallow features to extract high-order semantic features further to enhance the representation of central pixels, mitigating the limitation of central pixel-based classification.
3) We propose a new semantically constrained loss function for aligning multimodal semantic features. Our proposed loss function enhances inter-class separability and reduces intra-class variation by imposing constraints on the semantic similarity between the two modalities.

The structure of this article is organized as follows. Section II comprehensively reviews the relevant literature on the joint classification of hyperspectral and LiDAR data. Section III describes the proposed deep joint classification network. Section IV presents the experimental results and analyzes the proposed method's performance compared to other state-of-the-art methods. Finally, Section V concludes the article and discusses possible future directions for research.

II. Related Work

A. Multimodal Image Feature Fusion Methods

Recent research [29], [30] has demonstrated that the fusion of HSI and LiDAR data can significantly enhance classification accuracy, with feature fusion being the primary method used in most studies. In addition to feature-level fusion, Hang et al. [31], [32], [33] proposed a method of decision-level fusion that fully exploits the ensemble properties of deep neural networks. Effective cross-modality interaction is crucial for optimal information fusion in hyperspectral and

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LiDAR data classification. Prior research [34] has shown that cross-modality interaction can improve content understanding across multiple modalities. Furthermore, traditional hyperspectral and LiDAR data classification algorithms often integrate feature interaction into the feature fusion process, as shown in previous work [35], [36], [37], [38], [39]. Recent experimental evidence suggests direct feature interaction can enhance certain features' abundance and improve classifiers' discriminative capacity. In recent studies, such as [20], [31], the weighting approach was used for feature interaction during feature extraction. In addition, [40] used LiDAR features as spatial attention for weighting HSI features. Zhao et al. [41] proposed to use octave convolution [42] to decompose low-frequency and high-frequency features and perform feature interaction between same-frequency features of the two branches. S^2ENet [43] is a network that employs a spatial–spectral enhancement module (S^2EM) to facilitate cross-modal information interaction. The IP-CNN proposed in [44] employs a bidirectional autoencoder to perform joint reconstruction of hyperspectral and LiDAR data without requiring annotated information. Multisource structural information is integrated by imposing the HSI-perception and LiDAR-perception constraints. A^3CLNN [45] proposes a novel dual-channel CNN with spatial, spectral, and multiscale attention and long short-term memory for the feature extraction and classification of multisource RS data. Sal^2RN [46] introduces a cross-layer interaction module to dynamically adjust the importance of features across different layers and integrate these various features. Zhao et al. [47] proposed a hierarchical random walk network (HRWN) to study the joint classification of HSI and LiDAR data. Deeper layers of CNNs benefit from hierarchical random walk layers guided by spatial constraints and local seeds. Random walk layers alleviate the problem of weak localization around boundaries and spatial fragmentation of classification maps.

The Transformer model gained attention from researchers due to its ability to model long sequence correlations. Proposed initially for natural language processing, the Transformer model has since been applied to image processing, and some research has used it for RS classification tasks [48]. Roy et al. [49] proposed a multimodal fusion transformer for the joint classification of HSI and LiDAR data. This method uses LiDAR data as a learnable token to facilitate feature learning alongside HSI data. However, this operation only partially integrates the valuable information from the two modalities, which limits the classification accuracy.

B. Semantic Enhancement Methods

In multimodal fusion, semantic enhancement technology enhances features by extracting the semantic representation of shallow features. Liu et al. [50] propose a novel approach that combines object-based post-classification refinement (OBPR) and CNNs for accurate land cover mapping using optical and SAR data from the Sentinel satellite system. Many researchers have applied semantic enhancement methods to different RS scenarios. Numerous researchers have used semantic augmentation techniques in various RS contexts. OMM-CNN [51] presents a multimodal deep CNN architecture capable of extracting multiscale spectral–spatial contextual information. The proposed approach integrates multiscale features and semantic information from both modalities in a deep neural network architecture, which enhances the accuracy and efficiency of building extraction. MultiOCNN [52] employed multiscale CNNs to capture contextual information for semantic labeling of objects. These techniques improve the representation of the core pixel by combining multimodal data and semantic characteristics using deep neural networks.

Furthermore, the combination of semantic enhancement technology and deep learning has many applications in other fields. Zhang et al. [53] propose a dense semantic enhancement module (DSEM), an adversarial module that improves cross-domain instance-level feature matching in region-free detectors. The work presented in [54] investigates the potential of incorporating semantic knowledge to improve few-shot object detection. A semantic enhancement module is proposed to augment the partial visual representation of new categories. In contrast, a semantic prototype comparison loss is defined to learn a more discriminative embedding space that can effectively distinguish features belonging to distinct classes. However, only some researchers have applied semantic enhancement technology to land cover classification tasks based on central pixels. Semantic enhancement techniques enhance the representation of the center pixel by adding contextual data from surrounding pixels. These techniques address the shortcomings of central pixel classification, such as approximate representation and inter-class confusion.

III. Methodology

A. Problem Formulation

In this article, the HSI is represented by $\mathbf{X^H} \in \mathbb{R}^{M\times N\times B}$, and the corresponding LiDAR image is represented by $\mathbf{X^L} \in \mathbb{R}^{M\times N}$, where M and N represent the size of the two modal images, respectively, and B represents the spectral bands of hyperspectral. All pixels are assigned into C land cover classes, denoted by $C = (y_1, y_2, \ldots, y_c)$. The classification method adopts the central pixel classification. Each patch is centered at pixel (i, j) and has size $(k \times k)$. Patch extraction is carried out during the preprocessing stage. The patch of hyperspectral $\mathbf{x}_m^H$ with $k \times k$ pixel is obtained by normalizing the HSI data $\mathbf{X^H}$. And the corresponding patch of LiDAR $\mathbf{x}_m^L$ is obtained from LiDAR data $\mathbf{X^L}$.

Multimodal fusion for land cover classification is to improve the network's ability to distinguish different ground objects by jointly learning different features of HSI and LiDAR images. To divide the dataset into training sets D^{train} and testing sets D^{test}, we select corresponding patches from both HSI and LiDAR images and input them into the network for training. The samples in each class for $\mathbf{X}^H$ and $\mathbf{X}^L$ can be represented as follows for training and testing:

$$\begin{aligned} D^{\text{train}} &= \left\{\left(\mathbf{x}_1^H, \mathbf{x}_1^L\right), \left(\mathbf{x}_2^H, \mathbf{x}_2^L\right), \ldots, \left(\mathbf{x}_{n_t}^H, \mathbf{x}_{n_t}^L\right)\right\} \\ D^{\text{test}} &= \left\{\left(\mathbf{x}_{n_t+1}^H, \mathbf{x}_{n_t+1}^L\right), \ldots, \left(\mathbf{x}_{n_t+n_v}^H, \mathbf{x}_{n_t+n_v}^L\right)\right\} \end{aligned} \quad (1)$$

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where $\mathbf{x}_m^H$ and $\mathbf{x}_m^L$ are the spectral–spatial cube and the corresponding spatial cube, and n_t is the number of training samples, n_v is the number of testing samples.

Our proposed method can be described in the following steps: first, we obtain multiscale features of hyperspectral patches and LiDAR patches through the HSI and LiDAR feature Learning network (Section III-B). We then extract semantic features of the two modalities (Section III-C) to enhance multiscale features and compute cross-modal semantic constraint loss (Section III-E). Finally, the final prediction results are obtained through a multilevel residual fusion network (MRFN) (Section III-D). The workflow is summarized in Algorithm 1.

Algorithm 1 CMSE Model for Classification of Hyperspectral and LiDAR Data

Input: Hyperspectral Data $\mathbf{X^H}$, LiDAR Data $\mathbf{X^L}$, Network Model Θ

Output: The Trained CMSE Model Θ

```
Load hyperspectral data X^H and LiDAR data X^L
Divide training set D^train and testing set D^test by eq. (1)
Load the Network Model Θ
/* Start Training */
for e ≤ Epoch do
  for x_m^H, x_m^L in D^train do
    Obtain multiscale features of hyperspectral H̃ and LiDAR L̃ by eq. (2)
    Obtain semantic feature tokens of hyperspectral S^H and LiDAR S^L by eq. (3)
    Semantic enhancement on multiscale features
    Compute the cross-modal semantic loss ℓ_GW by eq. (10)
    Obtain fusion results through Multilevel residual fusion network
    Calculate the classification loss ℓ_CE by eq. (11)
    ℓ_total = αℓ_CE + βℓ_GW
    Update Network Θ Gradients
  end for
end for
/* Complete Training */
Save the Network Model Θ
Test the classification accuracy on the testing set D^test
```

B. HSI and LiDAR Feature Learning

Central pixel classification is often used in landcover classification tasks in RS because it offers low model complexity and high computational efficiency. The input to the network is a neighborhood of pixels. Due to differences in sensor characteristics, each RS information source data has significant differences in the original pixel-level space. Hyperspectral data usually contains hundreds of channels, while LiDAR data only has one. Therefore, it is essential to carefully consider the network architecture to utilize the information from both data sources effectively. We propose constructing the HSI and LiDAR feature learning network to extract and process the features of the neighborhood. In this article, we utilize a CNN as a feature extractor to transform the raw pixel-level data into feature-level data, effectively reducing spatial variance in the dataset. In order to balance spectral preservation and LiDAR redundancy, we adopt a network structure for shallow feature extraction, designing a network with varying numbers of channels in the HSI and LiDAR branches. In particular, the HSI channel count progressively decreases from 128 to 64, while the LiDAR channel count progressively increases from 8 to 64. By adjusting the width of the network, we can avoid losing valuable spectral information in the HSI branch while also preventing excessive redundancy in the LiDAR branch.

Center pixel classification is limited by its inability to effectively capture contextual information, which can lead to confusion between similar classes. In order to overcome this limitation, it is possible to utilize a multiscale approach for feature extraction to capture and utilize more contextual information. In this article, we propose a multibranch feature extraction (MBFE) block to overcome the limitations of single-scale feature extraction. The proposed MBFE block takes the feature maps of each modality as input and extracts sparse local features. Specifically, the MBFE block comprises convolution kernels of varying sizes, including 1×1, 3×3, and 5×5. Considering the hyperspectral input features map $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_C] \in \mathbb{R}^{C\times H\times W}$, the corresponding output feature map $\tilde{\mathbf{H}} = [\tilde{h}_{(s,1)}, \tilde{h}_{(s,2)}, \ldots, \tilde{h}_{(s,d_h)}] \in \mathbb{R}^{d_h\times H\times W}$ with the convolution kernel $\mathbf{V_s} = [\mathbf{v}_{(s,1)}, \mathbf{v}_{(s,2)}, \ldots, \mathbf{v}_{(s,d_h)}] \in R^{d_h\times C\times s\times s}$, which has a kernel size of $s \times s$, is calculated by

$$\tilde{h}_{(s,i)} = \mathbf{v}_{(s,i)} * \mathbf{H} = \sum_{k=1}^{C} \mathbf{v}_{(s,i)}^{[k]} * \mathbf{h}_k \tag{2}$$

where $*$ denotes the convolution operation, d_h is a channel-shrink factor, and $\mathbf{v}_{(s,i)}^{[k]}$ indicates the kth kernel parameters. Similarly, considering the LiDAR input feature map $\mathbf{L} = [\mathbf{l}_1, \mathbf{l}_2, \ldots, \mathbf{l}_C] \in \mathbb{R}^{C\times H\times W}$, we can obtain the convolution output $\tilde{\mathbf{L}} = [\tilde{l}_{(s,1)}, \tilde{l}_{(s,2)}, \ldots, \tilde{l}_{(s,d_h)}] \in \mathbb{R}^{d_h\times H\times W}$ with convolution kernel $\mathbf{U_s} = [\mathbf{u}_{(s,1)}, \mathbf{u}_{(s,2)}, \ldots, \mathbf{u}_{(s,d_h)}] \in R^{s\times s\times d_h\times C}$. Using this MBFE approach, our model can extract a wider range of diverse and discriminative features from the input data, improving classification accuracy.

C. Semantic Enhancement Based on Gaussian-Weighted Matrix

Recognizing that the exclusive use of the central pixel to represent entire objects may not always result in accurate classification outcomes, especially for detailed or complex objects, the limitations of the central pixel classification approach became apparent. It was considered that using only the center pixel classification method may lead to confusion between similar categories. Additionally, although the features obtained from the initial convolution operation encompass significant spectral and spatial data, they may only partially portray the distinctive attributes that differentiate various land cover constituents. To counteract this predicament, we endeavored to reinforce the feature representations by integrating a Gaussian-weighted category matrix. This matrix is intentionally crafted to capture the intricate semantics of HSI

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and LiDAR feature categories, facilitating more thorough and distinctive feature extraction.

For this part, the input multibranch HSI features are $\tilde{\mathbf{H}} = [\widetilde{h}_{(s,1)}, \widetilde{h}_{(s,2)}, \ldots, \widetilde{h}_{(s,d_h)}] \in \mathbb{R}^{H\times W\times d_h}$. Define its flattened feature map as $H_x \in \mathbb{R}^{uv\times z}$, where u is the height, v is the width, and z is the number of channels. For the flattened feature map of HSI H_x, semantic feature tokens S^H can be obtained by the following formula:

$$S^H = \underbrace{\text{softmax}(H_x \boldsymbol{W}_H)^T}_{\boldsymbol{A}} H_x. \tag{3}$$

In this equation, $\boldsymbol{W}_H$ denotes a weight matrix that is initialized using a Gaussian distribution. The operation $H_x \boldsymbol{W}_H$ indicates that a 1×1 filter performs a pointwise product. The objective is to project H_x onto the semantic subspace, resulting in a semantic representation denoted as $\boldsymbol{A}$. Subsequently, a transpose operation is performed on $\boldsymbol{A}$, followed by applying the softmax activation function to emphasize the salient semantic components. Finally, $\boldsymbol{A}$ multiplies with Hx to get the semantic enhancement feature S^H. Simultaneously, the input multibranch LiDAR features are $\tilde{\mathbf{L}} = [\widetilde{l}_{(s,1)}, \widetilde{l}_{(s,2)}, \ldots, \widetilde{l}_{(s,d_h)}] \in \mathbb{R}^{H\times W\times d_h}$. We can get three semantic enhancement feature S^{L1}, S^{L2}, S^{L3} of different sizes in a similar way. Semantically enhanced features derived from Gaussian weighting not only contain spatial and spectral information but also augment the semantic features of land cover, thereby improving their discriminability and robustness.

D. Multilevel Residual Fusion Network

In the context of joint land cover classification using LiDAR and HSI data, we recognize the importance of effectively using spatial information from LiDAR and hyperspectral data. Direct weighting or summation of features from both modalities may only partially exploit the spatial correlation between them, which may limit the performance of the classification system. To address this limitation and enhance spatial information integration, we propose a novel MRFN in this article. The MRFN consists of multiple fusion modules, which combine the features from both modalities at various levels. This network architecture enables the capture of intricate inter-modality relationships, thereby facilitating the extraction of highly discriminative features. Let the enhanced signal from HSI $S^H \in \mathbb{R}^{H\times W\times C}$ be x, we propose a general spatial enhancement operation as follows:

$$z_i = \text{Squeeze}\left(f\left(x_i, S_0^{L1}\right), \ldots, f\left(x_i, S_{N_{S^{L1}}-1}^{L1}\right)\right). \tag{4}$$

In this context, we define $S^{L1} \in \mathbb{R}^{H\times W\times C}$ as the LiDAR-enhanced signal. Moreover, $x_i \in \mathbb{R}^{C\times 1}$ and $S_j^{L1} \in \mathbb{R}^{C\times 1}$ denote the feature vectors at position i of x and position j of S^{L1}. The variable $N_{S^{L1}}$ represents the size of S^{L1} ($N_{S^{L1}} = H \times W$). A global relationship vector is aggregated into a scalar value by the operation denoted by Squeeze. The scalar z_i represents the spatial enhancement effect of all positions of S^{L1} on the ith position of x. Following the approach introduced by Wang et al. [55], the pairwise function f is used to compute the relationship between two positions

$$f(v_1, v_2) = \text{Transfor}_1(v_1)^T \text{Transfor}_2(v_2). \tag{5}$$

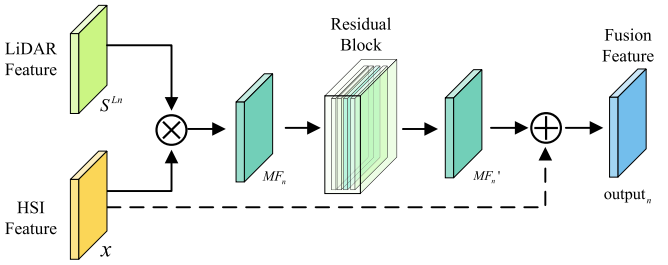

Fig. 2. Each level of residual fusion for HSI and LiDAR feature.

In this equation, v_1 and v_2 represent two vectors, while Transfor_1 and Transfor_2 represent transformation operations. In this article, we use the dot product as a function f. We also use a trainable 1×1 convolutional layer as a function Transfor. Based on (4), each position in the HSI feature map is associated with a ratio value. All positions of the LiDAR features enhance this ratio value. Finally, the transformed spatial enhancement map z is used for weighting x

$$\text{MF}_1 = x \otimes z. \tag{6}$$

Here, $\otimes$ denotes the element-wise multiplication performed on the spectral information from the HSI features and the spatial enhancement map z obtained from the LiDAR features, which results in a refined HSI feature map. MF_1 represents the feature enhanced by the fusion of HSI features using LiDAR features.

The previously obtained LiDAR features of different scales contain height features of different contexts, which can enhance HSI features to varying degrees. In this article, a multilevel residual module is employed for the fusion of LiDAR features and HSI features. MF_n' denotes the MF_n features passed through the residual module, and the value of n is 1, 2, 3. MF_n' is added to x to obtain the feature output_n of each level of residual fusion. Fig. 2 shows the process of each level of fusion. The resulting output from the residual fusion operation is utilized as the input x for the subsequent fusion level, thus completing the three-level fusion process to obtain the ultimate classification feature. Multilevel residual fusion networks enable the propagation of high-level features across multiple levels and reduce overfitting.

E. Cross-Modal Semantic Constraint Loss

In multimodal learning with LiDAR and HSI data, ensuring semantic consistency between corresponding ground objects in different modalities is paramount. However, achieving this consistency can be challenging due to the inherent differences in data representations and characteristics of each modality. To address this challenge and enhance the semantic consistency of multimodal features, we propose a novel cross-modal semantic constraint loss based on the Wasserstein distance [56], [57], [58]. The key objective of our proposed loss function is to penalize any semantic discrepancies between corresponding ground objects in LiDAR and HSI data and encourage the model to learn a unified and consistent semantic representation. By incorporating this loss function into the training process, the model can effectively learn to make semantically consistent predictions across modalities, thereby improving the performance and accuracy of multimodal learning tasks. Our proposed loss function utilizes the Wasserstein

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TABLE I
TRAINING SET AND TEST SET SAMPLE CATEGORY AND NUMBER OF HOUSTON, TRENTO, MUUFL DATASET

No.	Houston 2013			Trento			MUUFL		
	Class	Train	Test	Class	Train	Test	Class	Train	Test
1	Healthy Grass	198	1053	Apple trees	129	4034	Trees	2325	20921
2	Stressed Grass	190	1064	Buildings	125	2903	Grass Pure	428	3842
3	Synthetic Grass	192	505	Ground	105	479	Grass Groundsurface	689	6193
4	Trees	188	1056	Wood	154	9123	Dirt And Sand	183	1643
5	Soil	186	1056	Vineyard	184	10501	Road Materials	669	6018
6	Water	182	143	Roads	122	3174	Water	47	419
7	Residential	196	1072				Buildings' Shadow	224	2009
8	Commercial	191	1053				Buildings	625	5615
9	Road	193	1059				Sidewalk	139	1246
10	Highway	191	1036				Yellow Curb	19	164
11	Railway	181	1054				ClothPanels	27	242
12	Parking Lot 1	192	1041						
13	Parking Lot 2	184	285						
14	Tennis Court	181	247						
15	Running Track	187	473						
Total	-	2832	12197	-	819	30214	-	5375	48312

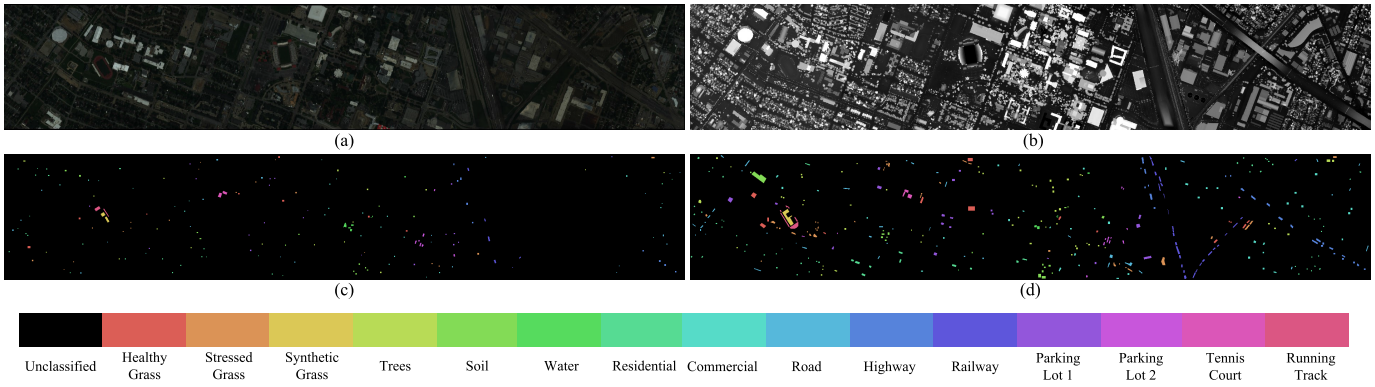

Fig. 3. Dataset visualization and classification maps for the Houston data (a) Hyperspectral pseudo-color imagery. (b) Grayscale image of LiDAR. (c) Training data map. (d) Testing data map.

distance $\mathbf{W}$ between two probability measures μ and ν, which correspond to the semantic vectors of two 2-D Gaussian distributions

$$\mathbf{W}(\mu;\nu) := \inf \mathbb{E}\left(\|\boldsymbol{W}_H - \boldsymbol{W}_L\|_2^2\right)^{1/2} \tag{7}$$

where $\boldsymbol{W}_H$ and $\boldsymbol{W}_L$ represent the semantic Gaussian matrix of HSI and LiDAR, respectively, with $\boldsymbol{W}_H \sim \mu$ and $\boldsymbol{W}_L \sim \nu$. We obtain the Wasserstein distance between two 2-D Gaussian distributions represented by μ and ν, respectively, as $d := \mathbf{W}(\mathcal{N}(\mathbf{m}_1, \boldsymbol{\Sigma}_1); \mathcal{N}(\mathbf{m}_2, \boldsymbol{\Sigma}_2))$. It can be written as

$$d^2 = \|\mathbf{m}_1 - \mathbf{m}_2\|_2^2 + \mathrm{Tr}\left(\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2 - 2\left(\boldsymbol{\Sigma}_1^{1/2}\boldsymbol{\Sigma}_2\boldsymbol{\Sigma}_1^{1/2}\right)^{1/2}\right) \tag{8}$$

where $\mathbf{m}_1$, $\boldsymbol{\Sigma}_1$ and $\mathbf{m}_2$, $\boldsymbol{\Sigma}_2$ denote the mean and covariance matrices of the two Gaussian distributions. They satisfy the positive definite symmetric matrix and can be exchanged with each other $\mathrm{Tr}(\Sigma_1\Sigma_2) = \mathrm{Tr}(\Sigma_2\Sigma_1)$, so the following equation holds:

$$\mathrm{Tr}\left(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2}\right) = \mathrm{Tr}(\Sigma_1\Sigma_2). \tag{9}$$

Therefore, cross-modal semantic constraint loss can be defined as

$$\ell_{\mathrm{GW}} = \|m_1 - m_2\|_2^2 + \left\|\Sigma_1^{1/2} - \Sigma_2^{1/2}\right\|_{\mathrm{Frobenius}}^2 \tag{10}$$

where $\|\cdot\|_{\mathrm{Frobenius}}^2$ represents the F-norm of the matrix. The cross-entropy loss ℓ_{CE} is computed between the fused features $v(m)$ and one-hot labels y_i. It can be expressed as follows:

$$\ell_{\mathrm{CE}} = \frac{-1}{N}\sum_{i=1}^{N}\left[\mathbf{y}_i \log \mathbf{v}_i^{(m)} + (1-\mathbf{y}_i)\log\left(1-\mathbf{v}_i^{(m)}\right)\right]. \tag{11}$$

Our proposed model updates its network parameters by jointly optimizing the following overall loss function:

$$\ell_{\mathrm{total}} = \alpha\ell_{\mathrm{CE}} + \beta\ell_{\mathrm{GW}} \tag{12}$$

where α and β are hyperparameters to adjust the weights of the ℓ_{CE} and ℓ_{GW}.

IV. EXPERIMENT

In this section, we perform extensive experiments on three popular image classification datasets, namely the Houston2013, the Trento, and the Muufl Gulfport datasets.

A. Datasets

1) Houston2013: The Houston2013 dataset was launched in 2013 by the IEEE Geoscience and RS Society (GRSS) Data Fusion Competition. The Houston2013 dataset was collected in 2012 in and around the University of Houston,

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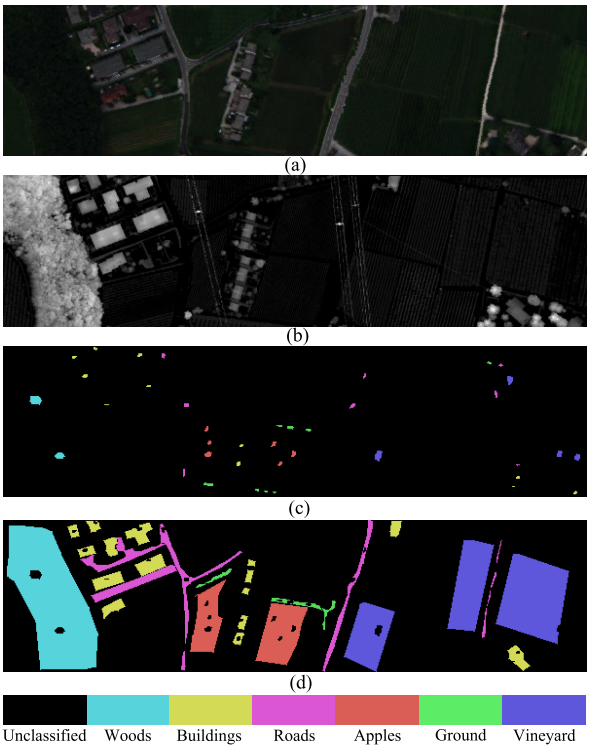

Fig. 4. Dataset visualization and classification maps for the Trento data (a) Hyperspectral pseudo-color imagery. (b) Grayscale image of LiDAR. (c) Training data map. (d) Testing data map.

Texas, with a spatial resolution of 2.5 m. It comprises a HSI of 144 spectral bands, spanning the wavelength range from 380 to 1050 nm. The dataset has dimensions of 349 × 1905 pixels. The Houston2013 dataset encompasses two versions: the original version and the cloud-free version. For our study, we utilized the cloud-free version of the dataset, which has been processed to remove cloud cover on the right side of the map, ensuring a clear view of the land cover in the data. The Houston2013 dataset consists of 15 categories with varying sample sizes, as shown in Table I. Fig. 3 presents a visualization of the hyperspectral and LiDAR data in the Houston dataset, highlighting the locations of pixels used for training and testing.

2) Trento: The Trento dataset used in this article was collected in the Trento region of Italy at 1-m resolution. There are 63 spectral bands from 420.89 to 989.09 nm in the HSIs in this dataset. The dataset contains six categories, and the corresponding information about the categories and the sample size are shown in Table I. The visualization of the hyperspectral and LiDAR data in the Trento dataset and the distribution of the training and test samples in the spatial domain are shown in Fig. 4.

3) MUUFL: The Muufl Gulfport dataset was collected on the University of Southern Mississippi campus in 2010. The Muufl Gulfport dataset consists of 325 × 220 pixels. There are 64 spectral bands in the HSI, except for eight noise bands. The dataset consists of eleven urban land cover classes. The distribution of randomly selected 10% samples from each class is shown in Table I. In addition, a visualization of the spatial distribution of the hyperspectral and LiDAR data and the training and test samples in the Muufl Gulfport dataset is presented in Fig. 5.

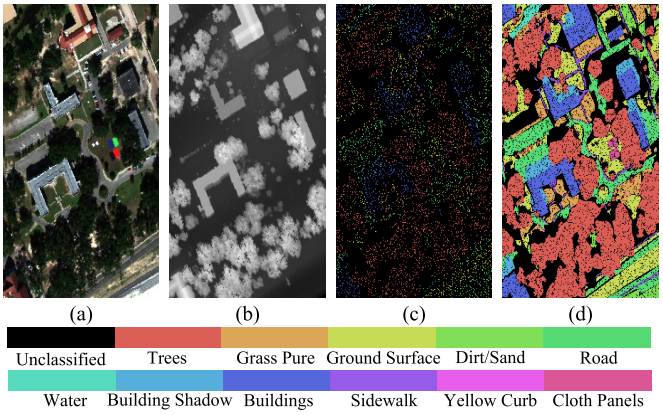

Fig. 5. Dataset visualization and classification maps for the MUUFL data. (a) Hyperspectral pseudo-color imagery. (b) Grayscale image of LiDAR. (c) Training data map. (d) Testing data map.

B. Experimental Setup

1) Implementation Details: Our proposed CMSE is implemented in the PyTorch programming platform. The raw HSI and LiDAR data are fed into the CMSE network, which is trained end-to-end. The input patch size for the network is set to 7 × 7 pixels. Training is performed for 150 epochs on a single 3080 GPU with batch size 32. A cosine annealing scheme adjusts the initial learning rate, set to 0.0001. The Adam optimizer is used in the training phase, with an initial learning rate of 0.001 and a decay factor of 0.5 after every three-tenths of the total epochs. The loss function of the CMSE network involves the parameters α and β, which are experimentally and empirically determined to be 0.4 and 0.6.

2) Evaluation Methods and Indicators: To evaluate the performance of our proposed CMSE model, we conducted both visual comparison and quantitative analysis. We have chosen representative deep learning-based approaches in hyperspectral and LiDAR data classification. These included Two-branch CNN [35], MDL-Early [20], MDL-Middle [20], MDL-Late [20], MDL-Cross [20], S^2ENet [43], EndNet [59], A^3CLNN [45], FusAtNet [40], IP-CNN [44], Sal^2RN [46], and BETF [60]. Two-branch CNN extracts features from hyperspectral and LiDAR images and splice them in a cascade manner. The early, middle, late, and cross-fusion models were proposed by [20], using CNN to extract two modal features and perform cross-attention fusion at different periods. S^2ENet uses a S^2EM, which enhances the features of the two modalities, respectively, and uses splicing to obtain the final features. FusAtNet uses LiDAR features as spatial attention for weighting HSI features. The IP-CNN employs a bidirectional autoencoder to perform joint reconstruction of hyperspectral and LiDAR data. A^3CLNN proposes a novel dual-channel CNN with spatial, spectral, and multiscale attention and long short-term memory for the feature extraction and classification of multisource RS data. Sal^2RN introduces a cross-layer interaction module to dynamically adjust the importance of features across different layers and integrate these various features. BTRF uses different Transformer encoders to extract multimodal spectral and spatial features and builds a binary

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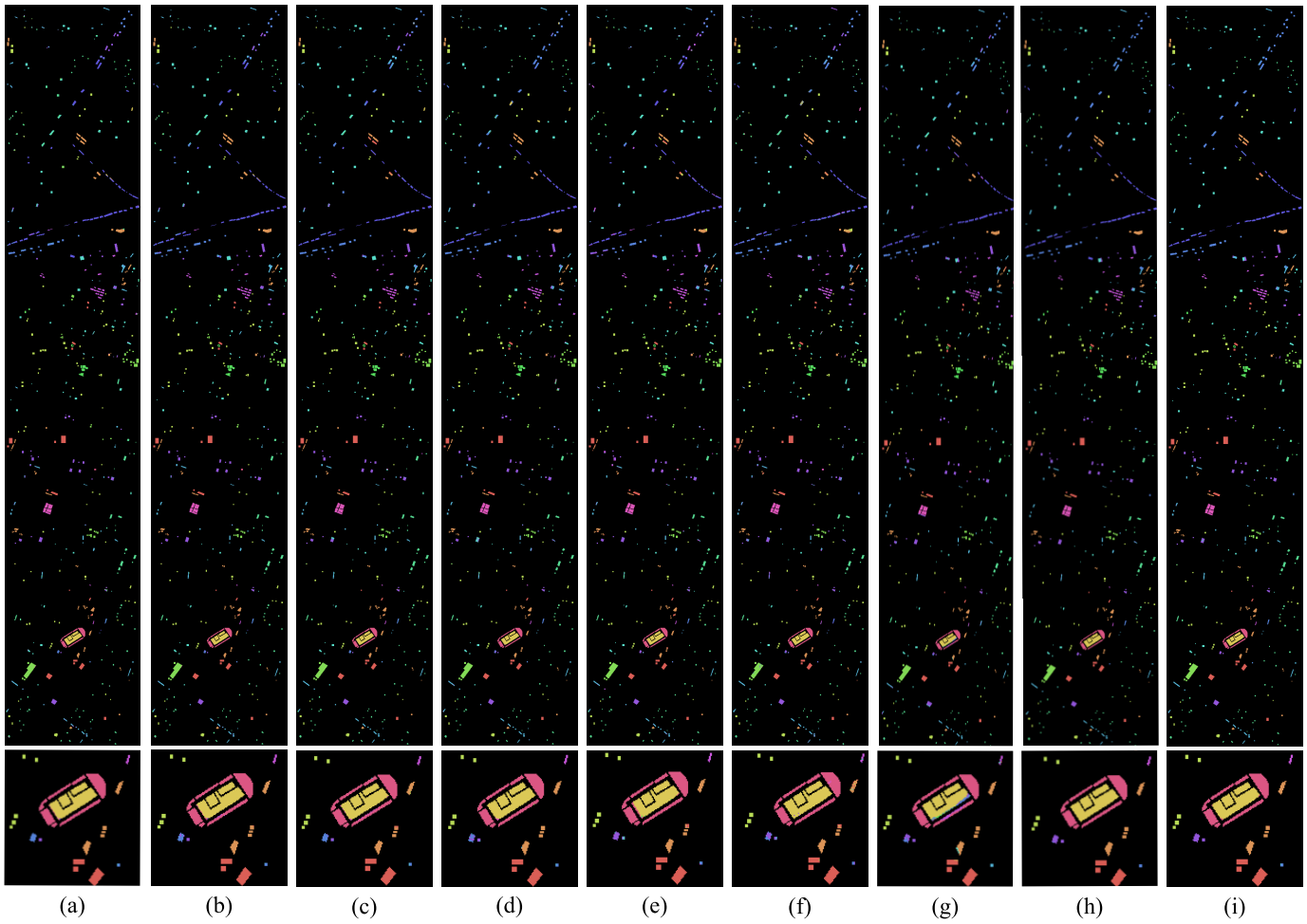

Fig. 6. Classification maps obtained using different approaches for the Houston dataset. (a) Two-branch (84.16%). (b) Early-fusion (87.18%). (c) Middle-fusion (89.55%). (d) Late-fusion (89.74%). (e) Cross-fusion (84.10%). (f) EndNet (88.52%). (g) FusAtNet (89.98%). (h) S^2ENet (94.19%). (i) Our (95.04%).

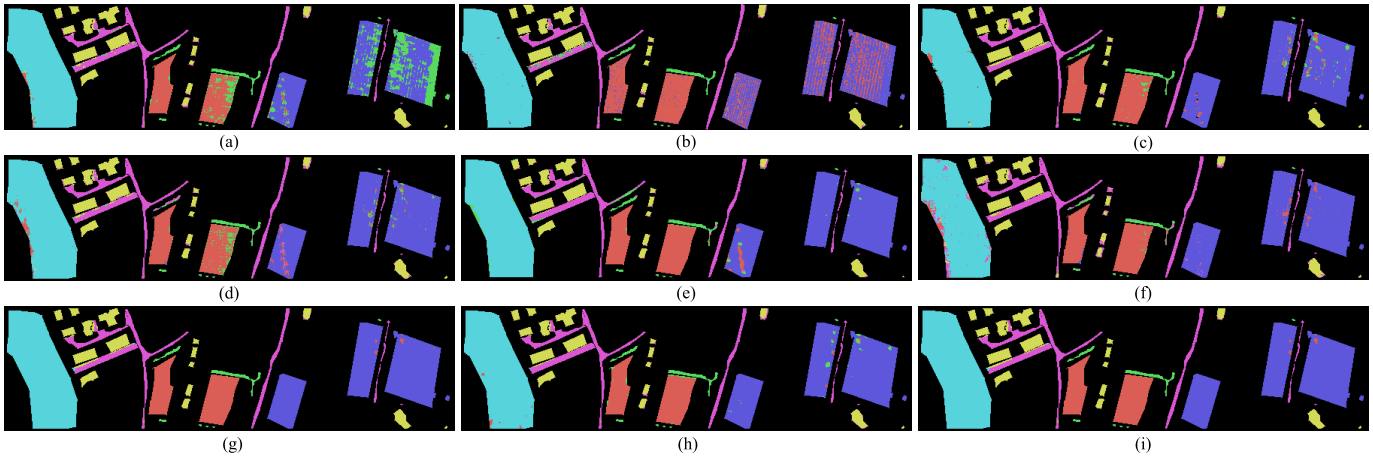

Fig. 7. Classification maps obtained using different approaches for the Trento dataset. (a) Two-branch (86.02%). (b) Early-fusion (86.92%). (c) Middle-fusion (96.09%). (d) Late-fusion (94.78%). (e) Cross-fusion (96.18%). (f) EndNet (94.17%). (g) FusAtNet (99.06%). (h) S^2ENet (97.83%). (i) Our (99.08%).

feature search tree to integrate multimodal features at different network levels. The visual comparison was conducted using sample classification maps, as shown in Figs. 6–8. For the quantitative evaluation, we have used several widely used metrics, such as the class accuracy (CA), the overall accuracy (OA), the average accuracy (AA), and the kappa coefficient. These metrics provide a comprehensive evaluation of the classification performance of our model. Higher values for these metrics generally indicate better classification accuracy in RS image classification tasks.

C. Experimental Results on Houston2013

The Houston2013 dataset comprises 2832 training and 12 197 test pixels, encompassing a wide range of land cover classes such as grasslands, trees, buildings, and roads, totaling 15 classes as outlined in Table I. Our experimental setup

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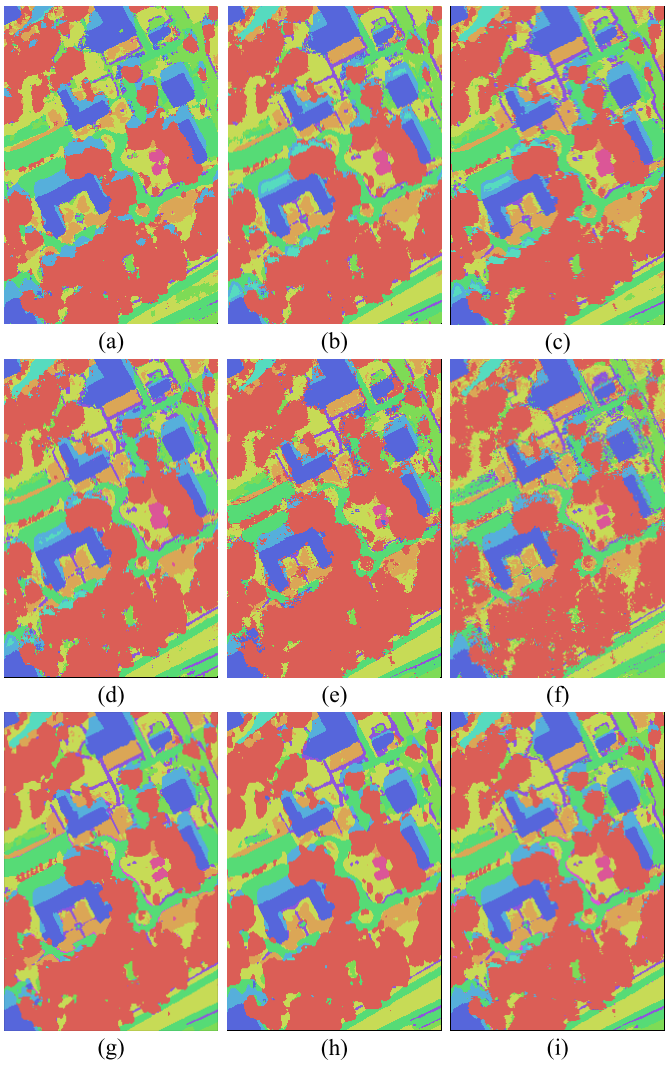

Fig. 8. Classification maps obtained using different approaches for the Muufl Gulfport dataset. (a) Two-branch (83.63%). (b) Early-fusion (87.44%). (c) Middle-fusion (87.94%). (d) Late-fusion (88.88%). (e) Cross-fusion (86.27%). (f) EndNet (89.69%). (g) FusAtNet (88.74%). (h) S^2ENet (89.70%). (i) Our (90.39%).

follows the protocol above. Table II presents the comparative performance analysis of various algorithms on the Houston dataset. The final row in Table II highlights the performance of our proposed CMSE network, which achieves the highest overall classification accuracy, average category accuracy, and Kappa coefficient compared to other methods on the Houston2013 dataset. Additionally, Fig. 6 displays the experimental results of our proposed method and other comparison methods, providing a closer look at some of the results.

The performance of several baseline deep learning networks, including Two-Branch CNN, Early-fusion, Middle-fusion, and Late-fusion, yielded overall classification accuracy of 84.16%, 87.18%, 89.55%, and 89.74%, respectively, indicating their limited effectiveness in fully leveraging the advantages of multisource feature fusion. Cross-fusion, which exchanges two modal feature information before fusion, achieved an overall classification accuracy of 84.1%, lower than other methods due to potential feature confusion caused by the interaction of similar feature information features in the presence of many classification categories. Endnet utilizes a codec structure to retain modal information but fails to fully consider the interaction between HSIs and LiDAR images, resulting in an overall classification accuracy of 88.52%. FusAtNet weights HSI features using LiDAR features and exhibits specific information interaction, achieving an overall classification accuracy of 89.98%. To facilitate cross-modal information interaction, S^2ENet uses a S^2EM. This interaction led to an overall classification accuracy of 94.19%. In comparison, the proposed CMSE network in this article guides the alignment of multimodal semantic information and considers the information of each modality. Residual fusion is employed to make up for the lack of spatial information in spectral features step by step, and multimodal information interaction is achieved, resulting in an overall classification accuracy of 95.04%, which significantly surpasses other methods.

Based on the individual CA, the CMSE network outperforms other methods in several categories, including Healthy Grass (C1), Soil (C5), Water (C6), Commercial (C8), Highway (C10), Railway (C11), and Running Track (C15). Due to the full consideration of the height information contained in LiDAR images, it has a good performance in the distinction between grassland (C1–C3) and trees (C4), which can solve the problem of foreign objects with the same spectrum in HSI classification. Considering the semantic information of each category, it has better classification performance on similar categories, such as roads (C9–C11). These results fully demonstrate the effectiveness of the proposed method.

D. Experimental Results on Trento

The Trento dataset consists of 819 training pixels and 30 214 test pixels, with six classes, including apple trees, buildings, roads, and others, as shown in Table I. The experimental framework is consistent with the one stated above. Table III illustrates the comparative analysis of classification accuracy of several algorithms on the Trento dataset. The performance of the proposed approach on the Trento dataset is shown in the last entry of Table III. The last three rows of the data in the Table III show that the proposed CMSE network outperforms the comparison methods regarding overall classification accuracy, average category accuracy, and the Kappa coefficient on the Trento dataset. In addition, Fig. 7 shows the experimental results of the proposed approach and the different comparison methods.

The performance of the baseline deep learning networks, two-branch CNN and early-fusion, on the Trento dataset yields an overall classification accuracy of 86.02% and 86.92%, respectively. Middle-fusion, Late-fusion, Cross-fusion, EndNet, and S^2ENet outperform the previous methods, indicating that increased interaction of multimodal features can enhance model performance, especially when dealing with a few feature categories. Notably, the Trento dataset has a concentrated distribution of similar features, with clear boundaries between different features. FusAtNet incorporates LiDAR features to weight HSI features, achieving an overall classification accuracy of 99.06%. The proposed CMSE network in this article guides the alignment of multimodal semantic information and fully considers the information of each modality. It utilizes

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TABLE II
CLASSIFICATION RESULTS FOR THE HOUSTON DATASET OBTAINED USING DIFFERENT APPROACHES

No.	Class	Two-branch	Early-fusion	Middle-fusion	Late-fusion	Cross-fusion	EndNet	FusAtNet	S^2ENet	A^3CLNN	IP-CNN	Sal^{2}RN	BFST	**Ours**
C1	Healthy grass	82.53	83.10	83.10	82.91	90.22	81.58	83.10	82.91	81.73	85.77	82.91	96.88	86.42
C2	Stressed grass	96.24	98.40	85.06	99.81	88.72	83.65	96.05	100.00	84.43	87.34	85.15	88.57	99.72
C3	Synthetic grass	97.82	97.82	99.60	99.01	93.86	100.00	100.00	100.00	91.49	100.00	93.27	100.00	98.61
C4	Trees	99.34	99.72	91.57	98.58	99.53	93.09	93.09	96.88	96.72	94.26	91.67	94.92	99.15
C5	Soil	99.15	99.34	98.86	99.91	99.34	99.91	99.43	99.91	99.97	98.42	100.00	98.10	100.00
C6	Water	96.50	97.90	100.00	97.90	95.80	95.10	100.00	100.00	97.90	99.91	100.00	99.15	100.00
C7	Residential	88.99	90.67	96.64	85.26	90.58	82.65	93.53	95.15	87.06	94.59	90.86	93.33	94.87
C8	Commercial	75.97	88.60	88.13	85.00	92.12	81.29	92.12	93.93	96.93	91.81	96.30	94.96	94.68
C9	Road	80.64	83.29	85.93	81.68	73.65	88.29	83.63	91.31	87.88	89.35	93.96	94.03	90.37
C10	Highway	34.85	43.05	74.42	84.36	27.99	89.00	64.09	92.05	70.82	72.43	83.59	90.18	96.43
C11	Railway	84.91	88.24	84.54	91.08	87.57	83.78	90.13	94.69	98.13	96.57	98.86	93.11	95.64
C12	Parking Lot 1	78.67	81.17	95.39	75.89	70.89	90.39	91.93	89.43	94.65	95.60	97.21	95.51	88.28
C13	Parking Lot 2	90.53	92.28	87.37	91.23	98.25	82.46	88.42	83.16	96.02	94.37	88.07	98.71	92.28
C14	Tennis Court	93.52	96.36	95.14	98.79	92.31	100.00	100.00	100.00	97.30	99.86	100.00	100.00	98.79
C15	Running Track	99.79	98.52	100.00	99.79	100.00	98.10	99.15	100.00	96.05	99.99	100.00	99.58	100.00
	OA(%)	84.16	87.18	89.55	89.74	84.10	88.52	89.98	94.19	90.55	92.06	93.45	94.47	**95.04**
	AA(%)	86.63	89.23	91.05	91.41	86.72	89.95	94.65	94.69	91.81	93.35	92.57	95.80	**95.68**
	Kappa	0.8282	0.8609	0.8871	0.8887	0.8279	0.8759	0.8913	0.9369	0.8975	0.9142	0.9193	0.9400	**0.9461**

TABLE III
CLASSIFICATION RESULTS FOR THE TRENTO DATASET OBTAINED USING DIFFERENT APPROACHES

No.	Class	Two-branch	Early-fusion	Middle-fusion	Late-fusion	Cross-fusion	EndNet	FusAtNet	S^2ENet	A^3CLNN	IP-CNN	Sal^{2}RN	BFST	**Ours**
C1	Apple trees	85.75	94.47	97.22	90.26	99.08	88.19	99.54	97.40	98.92	99.00	99.51	95.92	99.65
C2	Buildings	93.80	96.87	96.42	94.73	91.11	98.49	98.49	94.80	99.14	99.40	97.48	98.04	97.76
C3	Ground	100.00	97.29	98.33	82.46	93.11	95.19	99.73	100.00	98.12	99.10	96.52	80.33	99.79
C4	Wood	98.68	99.20	98.55	97.35	97.27	99.30	100.00	99.17	100.00	99.92	99.99	99.83	99.42
C5	Vineyard	69.84	69.48	94.42	95.02	96.67	91.96	99.90	97.94	99.95	99.66	100.00	99.92	99.90
C6	Roads	94.30	89.10	92.44	94.30	92.85	90.14	93.32	96.60	90.57	90.21	91.76	97.66	95.75
	OA(%)	86.02	86.92	96.09	94.78	96.18	94.17	99.06	97.83	98.73	98.58	98.80	98.66	**99.08**
	AA(%)	90.39	91.07	96.23	92.35	95.02	93.88	98.50	97.65	97.78	97.88	97.54	95.30	**98.71**
	Kappa	0.8225	0.8304	0.9483	0.9311	0.9495	0.9222	0.9875	0.9712	0.9831	0.9817	0.9839	0.9820	**0.9877**

TABLE IV
CLASSIFICATION RESULTS FOR THE MUUFL DATASET OBTAINED USING DIFFERENT APPROACHES

No.	Class	Two-branch	Early-fusion	Middle-fusion	Late-fusion	Cross-fusion	EndNet	FusAtNet	S^2ENet	**Ours**
C1	Trees	90.46	93.10	93.27	93.58	96.01	96.17	92.09	94.62	93.46
C2	Grass Pure	64.16	76.68	85.42	85.03	71.03	84.36	89.80	76.91	85.68
C3	Grass Groundsurface	73.78	82.35	74.57	79.77	78.91	75.17	84.16	89.70	86.82
C4	Dirt/Sand	77.36	72.43	76.51	71.94	75.17	85.82	69.45	69.63	76.45
C5	Road	92.29	92.14	92.29	91.82	83.00	92.85	89.86	93.64	92.09
C6	Water	10.98	85.68	89.74	89.26	69.21	86.87	82.58	65.16	89.26
C7	Building Shadow	89.20	70.23	78.85	85.76	70.38	77.45	91.44	89.10	94.87
C8	Buildings	92.57	92.57	92.24	92.70	92.82	95.23	89.92	93.21	92.16
C9	Sidewalk	30.18	63.32	71.43	71.19	53.13	56.82	69.66	59.71	70.55
C10	Yellow Curb	0.00	4.27	0.00	15.24	0.00	70.12	48.78	0.00	52.44
C11	Cloth Panels	86.36	92.98	96.28	97.93	70.25	95.45	88.84	87.60	94.21
	OA(%)	83.63	87.44	87.94	88.88	86.27	89.69	88.74	89.70	**90.39**
	AA(%)	64.30	75.07	77.33	79.48	69.08	83.30	81.51	74.48	**84.36**
	Kappa	0.7876	0.8366	0.8433	0.8553	0.8165	0.8637	0.8544	0.8653	**0.8750**

residual fusion to compensate for the lack of spatial information in spectral features, achieving multimodal information interaction. The overall classification accuracy of 99.08% outperforms other methods on the Trento dataset.

E. Experimental Results on MUUFL

The MUUFL dataset comprises 11 land-cover categories that are densely distributed, with some categories needing more samples, thereby posing challenges in their identification. Table IV presents the comparative performance analysis of various algorithms on the MUUFL dataset. The category of Yellow Curb (C10) poses a significant challenge due to its limited number of training pixels (only 19), and the accuracy

TABLE V
CLASSIFICATION RESULTS ACHIEVED BY DIFFERENT PATCH SIZE FOR THE HOUSTON, TRENTO, AND MUUFL DATASET

Datasets	Metric	5×5	7×7	9×9	11×11
Houston	OA(%)	94.87	95.04	90.83	89.41
	AA(%)	95.61	95.68	91.18	89.54
	Kappa	0.9444	0.9461	0.9004	0.8850
Trento	OA(%)	98.19	99.08	97.42	95.89
	AA(%)	96.05	98.71	94.93	92.97
	Kappa	0.9759	0.9877	0.9657	0.9453
MUUFL	OA(%)	89.41	90.39	88.36	86.38
	AA(%)	81.16	84.36	79.54	77.67
	Kappa	0.8624	0.8750	0.8494	0.8246

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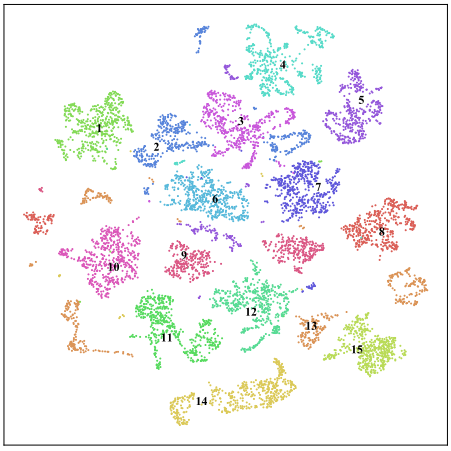
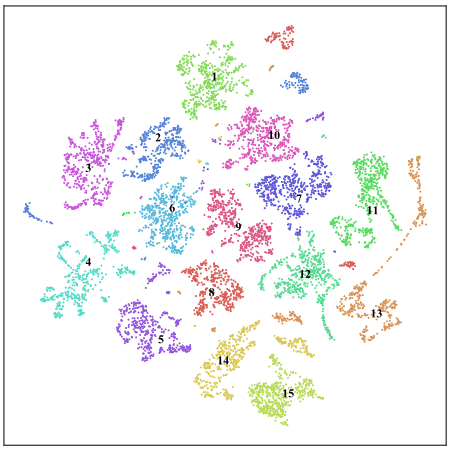
TABLE VI
IMPACT OF TRAINING SET SIZE ON CLASSIFICATION ACCURACY OF PROPOSED METHOD

Dataset	Indicator	Rate of Training Set					
		10%	20%	40%	70%	90%	100%
Houston2013	OA(%)	88.72	92.10	93.29	93.66	94.88	95.04
	AA(%)	86.71	87.15	0.8857	94.64	95.42	95.68
	Kappa	0.8777	0.9143	0.9272	0.9312	0.9445	0.9461
Trento	OA(%)	95.36	95.88	96.62	97.41	98.69	99.08
	AA(%)	91.91	92.97	96.73	96.66	98.11	98.71
	Kappa	0.9384	0.9453	0.9553	0.9656	0.9826	0.9877
MUUFL	OA(%)	81.46	84.75	87.52	88.53	89.46	90.39
	AA(%)	71.19	74.66	80.13	81.90	82.30	84.36
	Kappa	0.7615	0.8027	0.8373	0.8505	0.8624	0.875



(a)



(b)

Fig. 9. Visualization of feature distribution. (a) Visualization on the model without semantic enhancement. (b) Visualization on model with semantic enhancement.

of this category heavily relies on the model's generalization ability. While several comparison methods exhibit low accuracy for this category, the proposed method achieves 52.44% accuracy. While the proposed method is not the most accurate for each category in the Muufl Gulfport dataset, there are no categories with notably low accuracy. Endnet leverages the codec structure to preserve modal information, yielding an overall classification accuracy of 89.69%. In contrast, the proposed method fully considers the information of each modality,

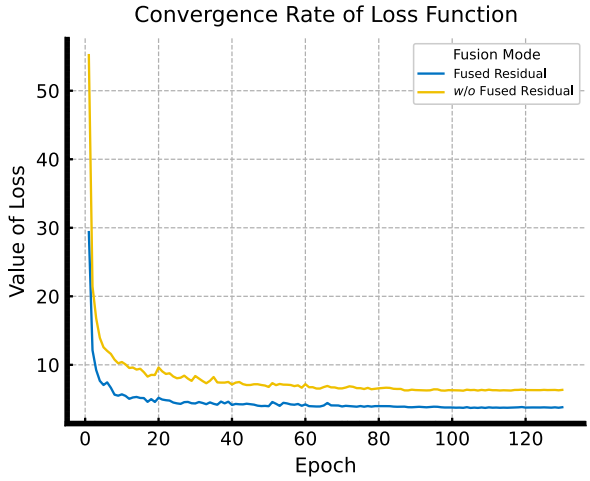


Fig. 10. Convergence rate of loss function with or without residual fusion module.

TABLE VII
CLASSIFICATION RESULTS ACHIEVED BY DIFFERENT LOSS FUNCTION FOR THE HOUSTON DATASET

Method	Metric		
	OA(%)	AA(%)	Kappa
Ours	**95.04**	**95.68**	**0.9461**
w/o ℓ_{GW}	93.29	94.47	0.9272
w/ ℓ_{KL}	93.93	94.59	0.9341
w/ ℓ_{JS}	94.02	94.60	0.9351

guiding the alignment of multimodal semantic information to classify objects in complex scenes effectively.

F. Parameter Tuning and Module Analysis

1) Multiscale Comparison: We then investigate the impact of varying input patch sizes on the model's performance. Table V presents the classification performance obtained using different window sizes, including 11×11, 9×9, 7×7, and 5×5. The results suggest that the choice of input patch size significantly affects the classification accuracy. Specifically, the 7×7 input patch size yields the best performance across all datasets. This finding is attributed to the fact that too small input patches can result in inadequate semantic information extraction. In contrast, too large input patches may lead to confusion in semantic information extraction. The optimal patch size of 7×7 is appropriate for subsequent semantic enhancement modules.

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TABLE VIII
MODEL COMPLEXITY AND OPERATIONAL EFFICIENCY OF METHODS ON HOUSTON 2013 DATASET

Method	ATT(s)	AIT(s)	Total Params	FLOPs
Two-Branch	273.5	72.4	75.783K	4.175M
MDL-Middle	301.7	78.4	121.214K	5.345M
MDL-Late	310.9	84.8	127.934K	5.385M
EndNet	168.8	30.6	91.093K	183.538K
S^2ENet	325.7	94.1	270.911K	32.804M
Ours	904.3	137.4	674.369 K	33.476M

2) Robustness Analysis: Table VI demonstrates the effect of training set size on the classification accuracy of the proposed method. We have observed that the OA, AA, and Kappa coefficient have generally increased as the proportion of the training set increases. This trend is expected since more labeled data can be used to train the classifier, resulting in better performance. For the Houston2013 dataset, our proposed method achieved an OA of 88.72% with only 10% of the data used for training. This accuracy increased by 6.32% when the entire dataset was used for training. For the Trento dataset, the OA increases by 3.72% as the size of the training set increases from 10% to 100%. Likewise, for the MUUFL dataset, the OA increases by 8.93% as the training set size increases from 10% to 100%. These findings imply that utilizing a more significant proportion of the training set is advantageous for our proposed method, allowing it to attain high classification accuracy, even with a small training set. However, performance improvement decreases as the training set grows, suggesting that the proposed method may achieve optimal performance with a specific quantity of labeled data.

3) Validity Analysis of Semantic Enhancement: To deeply understand why the proposed algorithm works, we visualize the features obtained by the UMAP algorithm in the Houston dataset, as shown in Fig. 9. Fig. 9(a) shows the result of feature clustering without semantic augmentation, and Fig. 9(b) shows the result of feature clustering with semantic augmentation. Points of different colors in the figure represent different categories. From Fig. 9(a), it can be seen that some category features are seriously confused when semantic enhancement is not used. For example, the feature distributions of categories 2, 9, and 13 in the feature space are split by features of other categories. When semantic enhancement is used, the feature distribution of each category is relatively concentrated, and almost no feature distribution is split, as shown in Fig. 9(b).

This article proposes a Gaussian weight-based semantic enhancement module to augment the representation of a central pixel by incorporating contextual information from the surrounding pixels. Fig. 9(a) and (b) demonstrate that semantic feature enhancement can effectively narrow the intraclass gap, improve the interclass gap, and provide a more distinct and stable feature space partition. Consequently, the proposed method has the potential to overcome the limitations of central pixel classification and reduce the potential for confusion among similar categories.

4) Influence Analysis of Fused Residual: Fig. 10 displays the overall loss curve, comparing the model's performance with and without the proposed residual fusion module as the number of training iterations increases. As the figure shows, adding the residual fusion module results in a relatively stable loss function and faster convergence. We propose a multilevel residual fusion module that gradually fuses spectral features in the HSI by leveraging elevation information captured by different branches of LiDAR images. The gradual fusion process compensates for missing spatial information in spectral features, allowing nonadjacent features to be supplemented. Inspired by the residual connection concept, a residual fusion module is proposed to enable the network to determine the redundancy layer, effectively utilizing multilevel features and reducing information redundancy.

5) Influence Analysis of Semantic Gaussian Wasserstein Loss: Table VII presents the impact of the proposed semantic Gaussian Wasserstein loss and other contrastive losses on classification accuracy. The compared methods include KS divergence and JS divergence. Without any cross-modal constraint loss, the overall classification accuracy is 93.29%, with an AA of 94.47%. Employing the proposed semantic Gaussian Wasserstein loss yielded an increase of 1.75% in the overall classification accuracy and 1.21% in the average classification accuracy. When adopting KS and JS divergence, the average classification accuracy increases by 0.64% and 0.73%, respectively.

KL and JS divergence are commonly used to measure the dissimilarity between probability distributions. In this article, we propose using them to quantify the semantic distribution difference between HSI and LiDAR. Moreover, if the semantic distribution of HSI and LiDAR is far apart, the JS divergence value becomes a constant, and the KL divergence value becomes infinite, which leads to vanishing gradients and prevents successful training. We introduce a novel loss function called semantic Gaussian Wasserstein loss, which utilizes the Wasserstein distance. The Gaussian Wasserstein loss is a suitable measure for evaluating the semantic distribution of two modes because it considers the structure between the distributions, not just their values.

6) Computational Analysis: To analyze the computational complexity of various models, we computed the total parameters, FLOPs, average training time (ATT), and average inference time (AIT) of the proposed and other state-of-the-art models. Table VIII presents the results. Notably, the proposed approach exhibits greater computational complexity than several other models listed in the table, leading to prolonged average training and inference times. However, it is essential to note that the proposed model demonstrates significantly improved OA and performance indicators. Additionally,

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we highlight that the model was trained and evaluated using specific hardware configurations; therefore, calculation requirements may differ on other hardware. We have included the number of parameters and FLOPs of the proposed method in the table to assist readers with understanding the model's computational complexity.

V. Conclusion

In this article, a CMSE network is proposed for HSI and LiDAR data fusion, which aims at mining and fusion of multimodal semantic information. Our experimental results on three multimodal datasets show that the proposed CMSE achieves significantly higher accuracies than other comparable approaches. Utilizing the multiscale deep feature learning module in CMSE, we can extract spatial and spectral features at different levels, which play a significant role in the classification process. Semantic enhancement and cross-modal semantic alignment make class features more discriminative. Multilevel residual fusion is used to make up for the insufficient elevation information in spectral features step by step and avoid feature redundancy. Therefore, our results demonstrate that the proposed CMSE can learn semantic features and conduct effective classification postprocessing, resulting in notable improvements in classification accuracy.

Using the percentage of labeled samples for classification in the MUUFL dataset leads to a class imbalance problem, which impacts classification performance. In future work, we prefer to focus on the imbalanced class problem for multimodal image classification.

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Wenqi Han is currently pursuing the Ph.D. degree with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China.

His research interests include remote-sensing image semantic segmentation and deep learning.

Wang Miao is currently pursuing the Ph.D. degree with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China.

His research interests include remote-sensing image scene classification and deep learning.

Jie Geng (Member, IEEE) received the B.S. degree in electronic information engineering and the Ph.D. degree in signal and information processing from the Dalian University of Technology, Dalian, China, in 2013 and 2018, respectively.

He is currently an Associate Professor at the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China. His research interests include SAR image processing, remote sensing image classification, and few-shot learning.

Wen Jiang received the bachelor's and master's degrees from Information Engineering University, Zhengzhou, China, in 1994 and 1997, respectively, and the Ph.D. degree from Northwestern Polytechnical University, Xi'an, China, in 2009.

She is currently a Professor at the School of Electronics and Information, Northwestern Polytechnical University. Her research interests include information fusion, artificial intelligence, remote sensing image processing and intelligent algorithm security.

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